ARC Approved-Solver Complexity, Human Difficulty, and LLM Difficulty

April 7, 2026

Overview

This note consolidates the work done on the 120 approved ARC Python solvers and the linked human/LLM psychometric data. The central question was whether solver complexity tracks task difficulty, and if so, whether it tracks *human* difficulty and *LLM* difficulty in the same way.

The short answer is that the results look **focused rather than random**:

- Humans and LLMs share a real difficulty axis.
- The **LLM-specific** part of difficulty is more aligned with **structural solver complexity**.
- The **human-specific** part of difficulty is more aligned with **time/search burden** and with **within-task heterogeneity across test pairs**.
- The earlier negative `thinking_advantage` result weakened materially after label auditing and floor diagnostics, so it should not be treated as a settled substantive claim.

Data

- Solver set: 120 approved ARC `solution.py` files, validated as correct.
- Complexity panel: all 120 approved files.
- LLM psychometric overlap: 56 approved eval rows (38 ARC-1 eval, 18 ARC-2 eval).
- Human-vs-LLM task overlap: 17 approved ARC-2 eval tasks.
- Human pair-level analysis: 110 public-eval task-pair rows with at least 8 human attempts.
- Multi-pair heterogeneity analysis: 90 pair rows across 44 tasks with multiple test pairs.

Methods and Null Hypotheses

Estimators.

- Pearson correlations for association claims
- Bootstrap 95% confidence intervals

- Permutation p-values for correlation tests
- Benjamini-Hochberg FDR correction across the main claim table
- Bootstrap differences of correlations for human-vs-LLM comparison claims
- Grouped binomial GLMs for `thinking_advantage`
- OLS residualization for shared-vs-specific analyses

Null hypotheses.

- Shared-axis null: human and LLM outcomes are unrelated across tasks.
- Difference-of-correlation null: a predictor is equally associated with human and LLM difficulty.
- Residual null: after removing the shared human/LLM axis, residual difficulty is unrelated to the proposed predictor.
- Human pair-level null: human pair difficulty is unrelated to the tested pair feature.
- Thinking-gap null: the thinking-vs-standard gap is unrelated to difficulty, or the GLM interaction is zero.

Complexity Result

The earliest robust signal was that **structural code size** correlates strongly with difficulty in this benchmark family.

- Best single structural proxy on the earlier latent-scale analysis: `ast_node_count`, $r = 0.666$
- Best conservative composite from PCR: leave-one-out $r = 0.691$
- On the expanded LLM analysis, `log1p(cyclomatic_complexity)` reached $r = 0.707$ against simple LLM logit difficulty

The interpretation is not "longer programs are universally harder problems." It is narrower: in a tightly standardized single-task domain like ARC, the amount of structured solver machinery needed to express a correct rule tracks task difficulty surprisingly well.

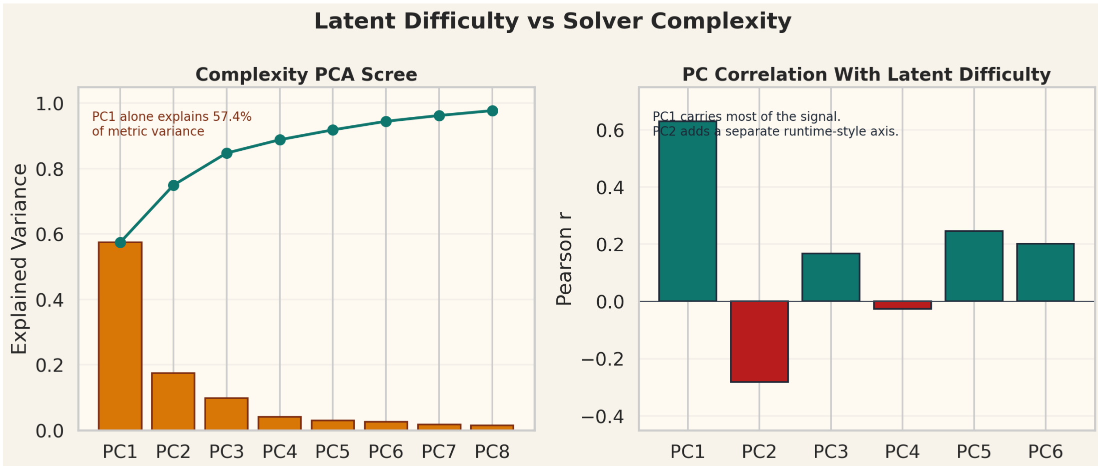

Figure 1: PCA view of the complexity panel. A large structural component dominates, with additional runtime and grid-size components.

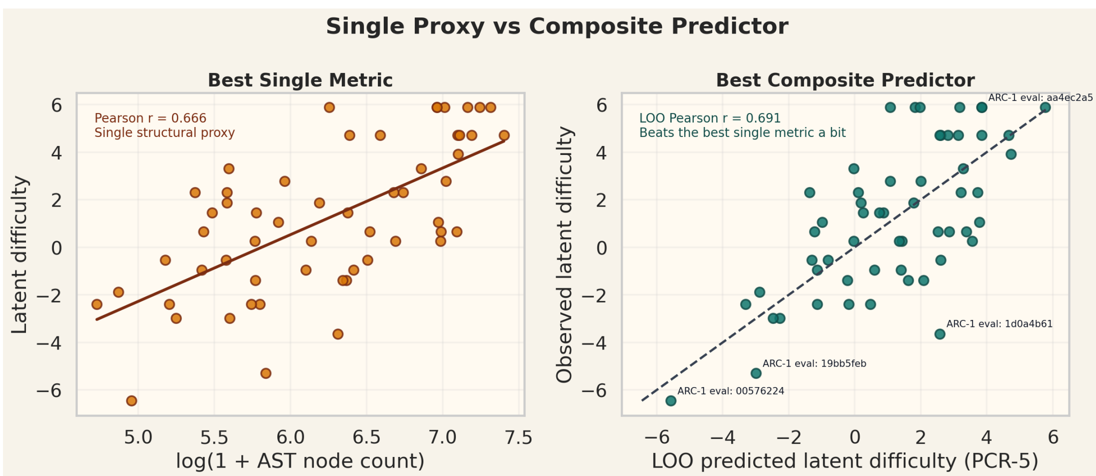

Figure 2: Best single complexity metric versus a conservative composite predictor. Structural metrics dominate runtime measures as predictors of difficulty.

Through-Line

The strongest through-line is:

1. Humans and LLMs share a meaningful difficulty axis.
2. Once that shared axis is removed, the leftover human difficulty and leftover LLM difficulty do *not* look the same.
3. LLM-specific residual difficulty still looks like solver structure.
4. Human-specific residual difficulty looks more like search/time burden, and pair-level human data show substantial within-task heterogeneity that task-level solver metrics cannot capture.

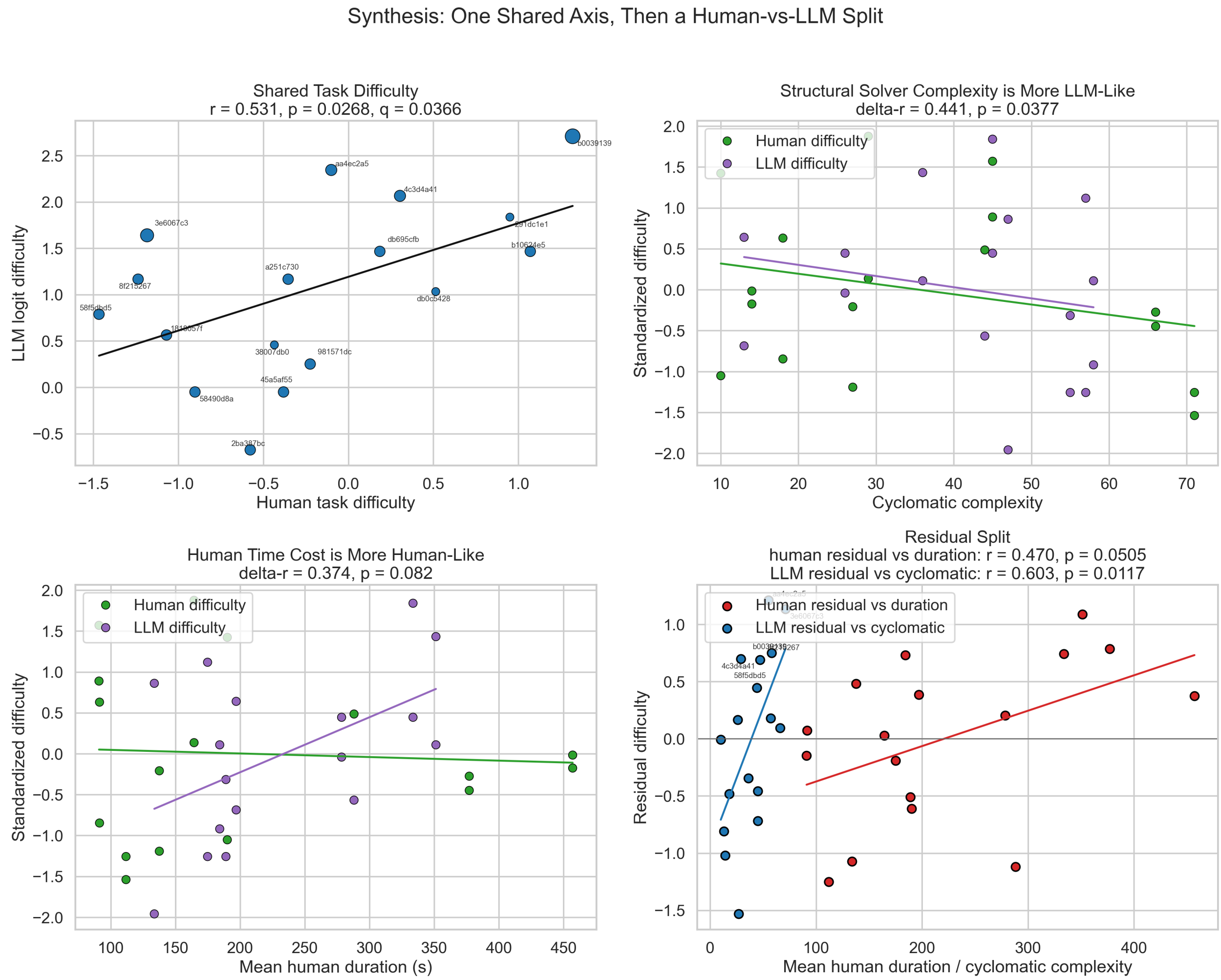

Figure 3: Main through-line figure: a shared human/LLM axis, followed by a human-vs-LLM split in the residual structure.

Strongest Supported Claims

Claim	Estimate	95% CI	p	q
Human difficulty aligns with LLM difficulty (S1)	$r = 0.531$	$[0.178, 0.780]$	0.0268	0.0366
Human solve rate aligns with LLM pass rate (S2)	$r = 0.541$	$[0.226, 0.781]$	0.0187	0.0280
Cyclomatic is more LLM-linked than human-linked (D1)	$\Delta r = 0.441$	$[0.024, 0.909]$	0.0378	0.0472
Residual LLM difficulty still tracks cyclomatic (D4)	$r = 0.603$	$[0.284, 0.816]$	0.0117	0.0194
Human pair difficulty tracks duration (H1)	$r = 0.391$	$[0.238, 0.539]$	1.67×10^{-4}	3.57×10^{-4}
Duration is more informative than board size (H3)	$\Delta r = 0.487$	$[0.255, 0.713]$	1.25×10^{-4}	3.57×10^{-4}
More test pairs shrink the human-over-LLM gap (H4)	$r = -0.366$	$[-0.539, -0.165]$	3.33×10^{-4}	6.25×10^{-4}
Task fixed effects explain pair-level difficulty variation (H5)	$R^2 = 0.749$	NA	7.93×10^{-5}	3.57×10^{-4}

Internal consistency checks. The older shared-model latent scale and the pooled Rasch scale are almost identical ($r = 0.991$), and pooled Rasch versus simple LLM logit difficulty is also near-identical ($r = 0.970$). This matters because it shows the LLM difficulty signal is not fragile to the exact psychometric construction.

Residual Analyses

Human residual after removing LLM difficulty.

- Model: human difficulty regressed on LLM difficulty
- Residual human difficulty versus mean human duration: $r = 0.470$
- 95% CI $[0.067, 0.766]$, $p = 0.0505$, $q = 0.0583$

This is suggestive but not strong enough to elevate to the main corrected claim set. It still points in a consistent direction: the **human-specific** part of difficulty looks more time/search-like.

LLM residual after removing human difficulty.

- Model: LLM difficulty regressed on human difficulty
- Residual LLM difficulty versus cyclomatic complexity: $r = 0.603$
- 95% CI $[0.284, 0.816]$, $p = 0.0117$, $q = 0.0194$

This is the cleanest evidence that solver structure carries an **LLM-specific** signal, not just a generic difficulty effect shared with humans.

Human Pair-Level Structure

The pair-level human data add an important layer that task-level solver complexity cannot see.

- Human pair difficulty tracks mean duration: $r = 0.391$

- Raw board size alone is weak for human difficulty: $r = -0.096$, $p = 0.311$
- More test pairs shrink the human-over-LLM gap: $r = -0.366$
- Mean within-task difficulty range across multi-pair tasks: 0.719
- Maximum within-task difficulty range: 2.498

This matters because a single task can contain very different human difficulty across its test pairs, while the solver-complexity analysis has only one program per task.

Figure 4: Human-side pair-level structure. Human difficulty is more time/search-like than board-size-like, and within-task heterogeneity is substantial.

Thinking-Advantage Audit

The original result looked strong under the legacy grouping:

- raw `thinking_advantage` versus difficulty: $r = -0.492$, $p = 1.67 \times 10^{-4}$
- legacy grouped-binomial GLM interaction: -0.718, $p = 5.70 \times 10^{-6}$

But the label audit found five low-certainty assignments that materially mattered:

- `QwQ-32B-Fireworks`
- `gemini-3-pro-preview`
- `gpt-5-pro-2025-10-06`
- `gpt-5-2-pro-2025-12-11-high`
- `gpt-5-2-pro-2025-12-11-medium`

Under the source-backed verified grouping:

- standard-group zero-success items: 35/56
- thinking-group zero-success items: 0/56
- both-zero items: 0/56

And the negative gap becomes unstable:

- all rows: raw $r = -0.378$, logit-gap $r = -0.294$
- standard-nonzero subset: raw $r = 0.280$, logit-gap $r = 0.091$
- both-groups-interior subset: raw $r = -0.009$, logit-gap $r = -0.156$

So the responsible interpretation is that the negative `thinking_advantage` result is an **audited, fragile finding**, not a settled substantive claim.

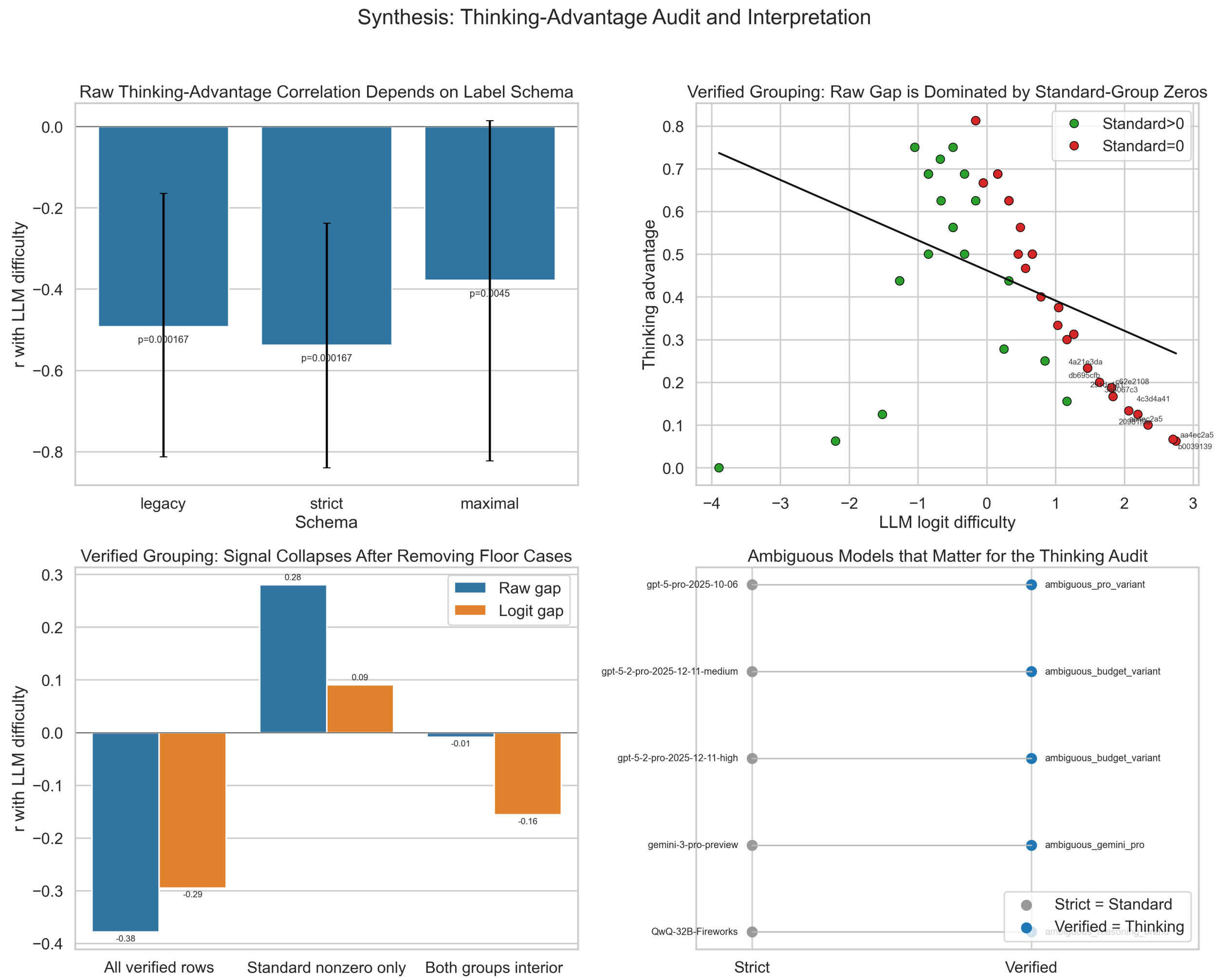

Figure 5: Thinking-advantage audit. The legacy negative result is not robust after source-backed relabeling and floor-sensitive subsetting.

What Did Not Clearly Survive

Claim	Estimate	95% CI	p	q
Duration: human > LLM (D2)	$\Delta r = 0.374$	$[-0.058, 0.803]$	0.0820	0.0879
Residual human ∼ duration (D3)	$r = 0.470$	$[0.067, 0.766]$	0.0505	0.0583
Board size predicts human difficulty (H2)	$r = -0.096$	$[-0.275, 0.086]$	0.3110	0.3110

Why This Looks Focused

If these were just disconnected correlations, I would expect them to scatter in unrelated directions. Instead, they line up into a coherent geometry:

1. Independent LLM difficulty constructions collapse onto one axis.
2. Human and LLM difficulty share a moderate common axis.

3. Structural solver metrics repeatedly favor the LLM side.
4. Human duration and pair-level heterogeneity repeatedly favor the human side.
5. The one attractive extra story, `thinking_advantage`, is exactly the result that weakened when the labeling and floor assumptions were stress-tested.

That last point is important: the pipeline did not simply preserve every interesting result. It also downgraded one when the diagnostics stopped supporting it.

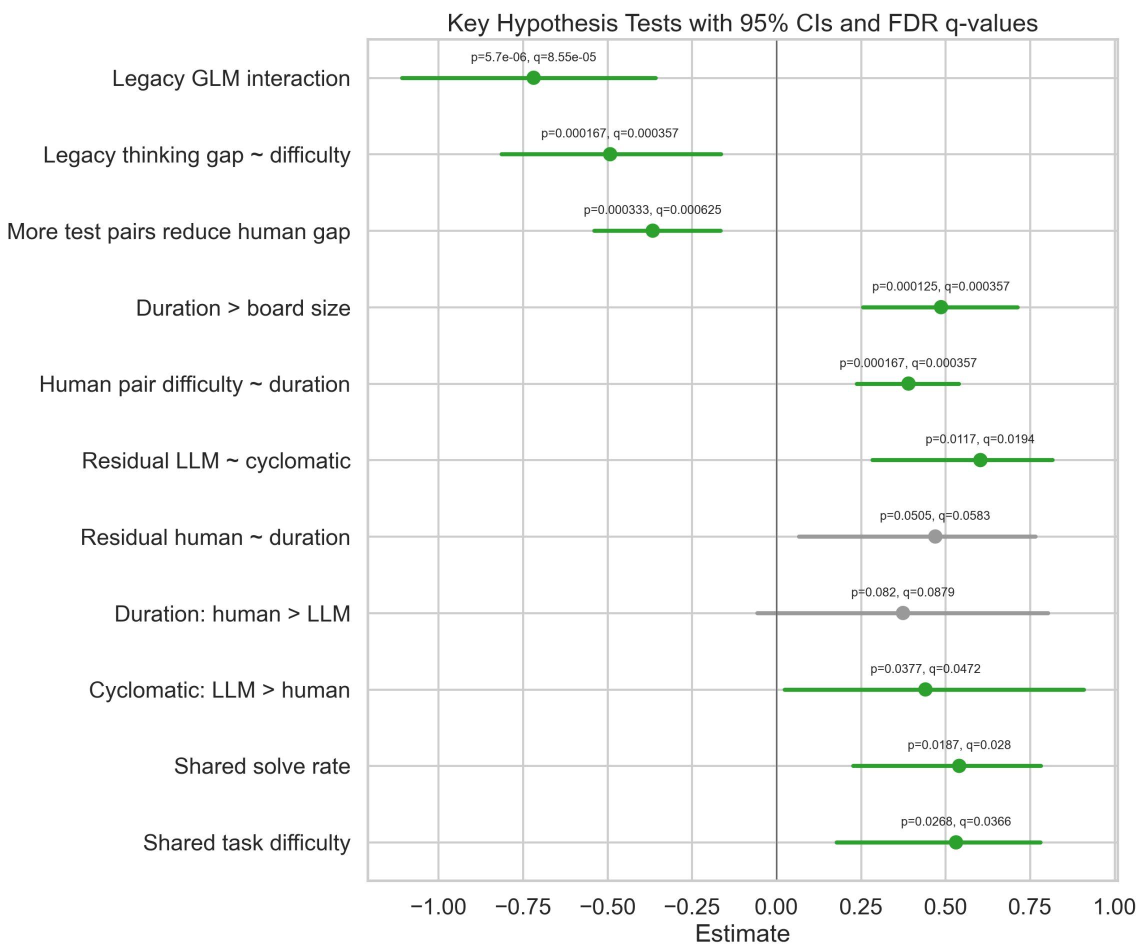

Figure 6: Key claims with confidence intervals and FDR-adjusted q-values.

Bottom Line

The most defensible statement from the full analysis is:

> In approved ARC solvers, structural program complexity is a strong proxy for LLM task difficulty, while human-specific difficulty looks more search/time-like and more sensitive to within-task heterogeneity. Humans and LLMs share part of the same difficulty axis, but not all of it.